EXPERIMENT 31 — PATTERN MATCHING

Aim

To write and execute a Prolog program to implement **pattern matching using unification**.

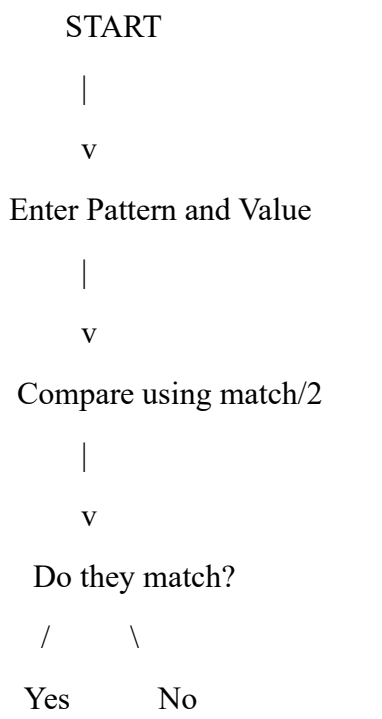
Objective

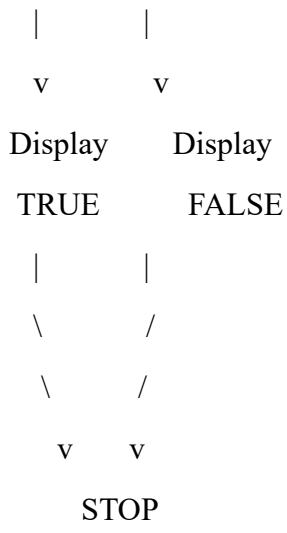
- To understand pattern matching in Prolog.
- To understand the concept of unification.
- To compare two patterns and determine whether they match.
- To retrieve values using variables.

Algorithm

1. Start the program.
2. Define the match/2 predicate.
3. Pass a pattern and a value as arguments.
4. Prolog compares both arguments using unification.
5. If both patterns match, return true.
6. If they do not match, return false.
7. If a variable is present, bind the variable to the matching value.
8. Stop.

Flowchart





Prolog Program

match(Pattern, Pattern).

This single rule performs pattern matching using **Prolog unification**.

Queries

Query 1

?- match(hello, hello).

Output

true.

Query 2

?- match(cat, dog).

Output

false.

Query 3

?- match(apple, X).

Output

X = apple.

Query 4

?- match(X, 100).

Output

X = 100.

Sample Output

?- match(hello, hello).

true.

?- match(cat, dog).

false.

?- match(apple, X).

X = apple.

?- match(X, 100).

X = 100.

Result

Thus, the Prolog program for **pattern matching using unification** was successfully implemented and executed.